Practical: 5.1

Problem Statement

Design and develop an Android application that displays an AppBar (Toolbar) at the top of the screen. The AppBar should show the application title and remain visible across the application interface to provide a consistent user experience. The title of the AppBar should be updated dynamically at runtime.

Project Configuration

Project Name: DynamicAppBarDemo

Language: Java

Activity Type: Empty Views Activity

Minimum SDK: API 21

Design Layout with AppBar

Open **res/layout/activity_main.xml**
Replace the entire code with the following:

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.coordinatorlayout.widget.CoordinatorLayout

    xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:app="http://schemas.android.com/apk/res-auto"

    android:layout_width="match_parent"

    android:layout_height="match_parent">

    <!-- AppBar -->

    <com.google.android.material.appbar.MaterialToolbar

        android:id="@+id/toolbar"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:background="?attr/colorPrimary"

        android:title="Dynamic AppBar"

        android:titleTextColor="@android:color/white"

        app:layout_scrollFlags="scroll|enterAlways"/>

    <!-- Main Content -->

    <LinearLayout

        android:orientation="vertical"
```

```
        android:padding="20dp"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:layout_marginTop="56dp">
        <EditText
            android:id="@+id/editTitle"
            android:hint="Enter new AppBar title"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>
        <Button
            android:id="@+id/btnChangeTitle"
            android:text="Change AppBar Title"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>
    </LinearLayout>
</androidx.coordinatorlayout.widget.CoordinatorLayout>
```

Explanation

MaterialToolbar → AppBar at the top

EditText → Accepts new title

Button → Updates AppBar title dynamically

AppBar remains **visible at all times**

STEP 3: Write Java Code

Open **MainActivity.java**

```
package com.example.dynamicappbardemo;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.widget.Toolbar;
public class MainActivity extends AppCompatActivity {
```

```
    Toolbar toolbar;
    EditText editTitle;
    Button btnChangeTitle;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        // Initialize AppBar
        toolbar = findViewById(R.id.toolbar);
        setSupportActionBar(toolbar);
        // Initialize UI components
        editTitle = findViewById(R.id.editTitle);
        btnChangeTitle = findViewById(R.id.btnChangeTitle);
        // Dynamic AppBar title change
        btnChangeTitle.setOnClickListener(v -> {
            String newTitle = editTitle.getText().toString();
            if (!newTitle.isEmpty()) {
                getSupportActionBar().setTitle(newTitle);
            }
        });
    }
}
```

Logic

setSupportActionBar(toolbar) converts Toolbar into AppBar

User enters text in EditText

Button click updates AppBar title dynamically

AppBar remains visible throughout the app

STEP 4: Run the Application

1. Click **Run**
2. App opens with AppBar showing default title
3. Enter a new title

4. Click **Change AppBar Title**
5. AppBar title updates dynamically

Prac:5.2

Aim To design and develop an Android Calculator Application that demonstrates the effective use of AppBar, Fragments, and basic UI components. The application aims to perform basic arithmetic operations based on user input handled within a fragment, while maintaining a consistent AppBar across the application interface to ensure a structured and user-friendly design.

Problem Statement

Design and develop an Android Calculator Application that demonstrates the use of AppBar, Fragments, and basic UI components. The application should perform arithmetic operations using user input handled inside a fragment, while maintaining a consistent AppBar across the application.

Project Details

Project Name: FragmentCalculatorApp

Language: Java

Activity Type: Empty Views Activity

Minimum SDK: API 21

STEP 1: Activity Layout with AppBar + Fragment Container

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.coordinatorlayout.widget.CoordinatorLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <!-- AppBar -->
    <com.google.android.material.appbar.MaterialToolbar
        android:id="@+id/toolbar"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:title="Fragment Calculator"
        android:background="?attr/colorPrimary"
        android:titleTextColor="@android:color/white" />
    <!-- Fragment Container -->
```

```
    <FrameLayout
        android:id="@+id/fragment_container"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:layout_marginTop="56dp"/>
</androidx.coordinatorlayout.widget.CoordinatorLayout>
```

Create Fragment

Right click **java → package**

New → Fragment → Fragment (Blank)

Name: fragment_calculator

Click **Finish**

STEP 2: Calculator Fragment Layout (UI Components)

fragment_calculator.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <EditText
        android:id="@+id/num1"
        android:hint="Enter first number"
        android:inputType="numberDecimal"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
    <EditText
        android:id="@+id/num2"
        android:hint="Enter second number"
        android:inputType="numberDecimal"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
```

```
<LinearLayout
    android:orientation="horizontal"
    android:layout_width="match_parent"
    android:layout_height="wrap_content">

    <Button
        android:id="@+id/btnAdd"
        android:text="+"
        android:layout_width="0dp"
        android:layout_weight="1"
        android:layout_height="wrap_content"/>
    <Button
        android:id="@+id/btnSub"
        android:text="-"
        android:layout_width="0dp"
        android:layout_weight="1"
        android:layout_height="wrap_content"/>
    <Button
        android:id="@+id/btnMul"
        android:text="×"
        android:layout_width="0dp"
        android:layout_weight="1"
        android:layout_height="wrap_content"/>
    <Button
        android:id="@+id/btnDiv"
        android:text="÷"
        android:layout_width="0dp"
        android:layout_weight="1"
        android:layout_height="wrap_content"/>
</LinearLayout>
<TextView
```

```
        android:id="@+id/txtResult"

        android:text="Result will appear here"

        android:textSize="18sp"

        android:paddingTop="20dp"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"/>

</LinearLayout>
```

STEP 3: Calculator Fragment Logic

CalculatorFragment.java

```
package com.example.fragmentcalculatorapp;
import android.os.Bundle;
import androidx.fragment.app.Fragment;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
public class CalculatorFragment extends Fragment {
    EditText num1, num2;
    TextView txtResult;
    public CalculatorFragment() {}
    @Override
    public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle
savedInstanceState)
    {

        View view = inflater.inflate(R.layout.fragment_calculator, container, false);
        num1 = view.findViewById(R.id.num1);
        num2 = view.findViewById(R.id.num2);
        txtResult = view.findViewById(R.id.txtResult);
        Button btnAdd = view.findViewById(R.id.btnAdd);
        Button btnSub = view.findViewById(R.id.btnSub);
        Button btnMul = view.findViewById(R.id.btnMul);
        Button btnDiv = view.findViewById(R.id.btnDiv);
        View.OnClickListener listener = v -> calculate(v.getId());
        btnAdd.setOnClickListener(listener);
        btnSub.setOnClickListener(listener);
        btnMul.setOnClickListener(listener);
        btnDiv.setOnClickListener(listener);

        return view;
    }
```

```
    private void calculate(int id) {
        double a, b;

        if (num1.getText().toString().isEmpty() ||
                num2.getText().toString().isEmpty()) {
            txtResult.setText("Please enter both numbers");
            return;
        }
        a = Double.parseDouble(num1.getText().toString());
        b = Double.parseDouble(num2.getText().toString());
        double result = 0;
        if (id == R.id.btnAdd) result = a + b;
        else if (id == R.id.btnSub) result = a - b;
        else if (id == R.id.btnMul) result = a * b;
        else if (id == R.id.btnDiv) {
            if (b == 0) {
                txtResult.setText("Division by zero not allowed");
                return;
            }
            result = a / b;
        }
        txtResult.setText("Result: " + result);
    }
}
```

STEP 4: Load Fragment in Activity

MainActivity.java

```
package com.example.fragmentcalculatorapp;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.widget.Toolbar;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Toolbar toolbar = findViewById(R.id.toolbar);
        setSupportActionBar(toolbar);
        if (savedInstanceState == null) {
            getSupportFragmentManager()
                    .beginTransaction()
                    .add(R.id.fragment_container, new CalculatorFragment())
                    .commit();
        }
    }
}
```